

Rigidity of Optimal Riesz Bounds for Induced Outer Products

Candidate full solution packet
arXiv:1410.7755, question after Theorem 4.6

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Status. Candidate full negative answer, pending expert review: no frames other than equiangular tight frames attain the simultaneous optimal pair. More strongly, equality in the sharp lower bound alone forces an equiangular tight frame.

1 Source question

Casazza, Pinkham, and Tuomanen [1] prove that the induced outer products of a unit-norm frame in $\mathbb{F}^N$ have upper Riesz bound at least M/N and lower Riesz bound at most $M(N-1)/(N(M-1))$. Equiangular tight frames attain both. They then ask whether any other frames can attain these optimal bounds:

We can think of equiangular tight frames as minimizers of the the quantity $B - A$ where A and B are the Riesz bounds of the induced outer products. One problem is that there are few equiangular tight frames. If we want to produce an outer product sequence with arbitrary size and dimension and have predictably good bounds, we cannot use equiangular tight frames. At this time we do not know if there are other frames which achieve the optimal bounds above.

Here $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, and $1 < N < M$. For unit vectors $\Phi = (\phi_i)_{i=1}^M \subset \mathbb{F}^N$, let

$$H = [|\langle \phi_i, \phi_j \rangle|^2]_{i,j=1}^M$$

be the Gram matrix of the rank-one operators $\phi_i \phi_i^*$. Write its eigenvalues as $\lambda_1 \geq \dots \geq \lambda_M$. When the outer products are a Riesz sequence, their optimal bounds are

$$B = \lambda_1, \quad A = \lambda_M.$$

2 Proof intuition

The source's lower-bound proof contains the whole rigidity mechanism. If the smallest eigenvalue reaches its upper limit, both the estimate on the largest eigenvalue and the averaging estimate on the other eigenvalues must be exact. Thus the outer-product Gram matrix has one eigenvalue of multiplicity one and one of multiplicity $M-1$. Its unit diagonal then fixes every coordinate of the exceptional eigenvector, while entrywise nonnegativity fixes their phases. This makes all off-diagonal entries equal. Finally, equality in the frame potential inequality forces tightness.

3 Full characterization

Set

$$\beta := \frac{M}{N}, \quad \alpha := \frac{M(N-1)}{N(M-1)}, \quad c := \frac{M-N}{N(M-1)}.$$

Theorem 1. *Let $\Phi = (\phi_i)_{i=1}^M$ be a unit-norm frame for $\mathbb{F}^N$, where $1 < N < M$, and suppose its induced outer products form a Riesz sequence with optimal bounds $A \leq B$. The following are equivalent:*

1. $A = \alpha$;
2. $(A, B) = (\alpha, \beta)$;
3. $B - A = \beta - \alpha$;
4. Φ is an equiangular tight frame.

In this case $|\langle \phi_i, \phi_j \rangle|^2 = c$ whenever $i \neq j$. Consequently, there are no other frames attaining the optimal pair asked about in [1].

Proof. The matrix H is positive semidefinite and has diagonal entries one, so $\text{tr } H = M$. Its entry sum is the frame potential. The standard frame-potential inequality and the Rayleigh quotient give

$$\frac{M^2}{N} \leq \mathbf{1}^* H \mathbf{1} \leq M\lambda_1, \quad \text{hence} \quad \lambda_1 \geq \beta.$$

Moreover,

$$(M-1)\lambda_M \leq \sum_{i=2}^M \lambda_i = M - \lambda_1 \leq M - \beta = (M-1)\alpha,$$

so $A = \lambda_M \leq \alpha$.

Assume first that $A = \alpha$. The last displayed chain has equal endpoints, so every inequality in it is equality. Therefore

$$\lambda_1 = \beta, \quad \lambda_2 = \cdots = \lambda_M = \alpha.$$

Since $\beta > \alpha$, the spectral theorem supplies a unit vector $u \in \mathbb{F}^M$ such that

$$H = \alpha I_M + (\beta - \alpha)uu^*.$$

Taking diagonal entries and simplifying gives, for every i ,

$$|u_i|^2 = \frac{1 - \alpha}{\beta - \alpha} = \frac{1}{M}.$$

For $i \neq j$, the number

$$H_{ij} = (\beta - \alpha)u_i \overline{u_j} = |\langle \phi_i, \phi_j \rangle|^2$$

is real and nonnegative, while its modulus is

$$|H_{ij}| = \frac{\beta - \alpha}{M} = \frac{M - N}{N(M - 1)} = c.$$

Thus $H_{ij} = c$ for all $i \neq j$, proving equiangularity.

It remains to prove tightness without assuming it. If S is the frame operator, then

$$\operatorname{tr} S = M, \quad \operatorname{tr}(S^2) = \sum_{i,j=1}^M |\langle \phi_i, \phi_j \rangle|^2 = M + M(M-1)c = \frac{M^2}{N}.$$

For the N nonnegative eigenvalues of S , Cauchy–Schwarz says $\operatorname{tr}(S^2) \geq (\operatorname{tr} S)^2/N$, with equality only when they are all M/N . Hence $S = (M/N)I_N$, and Φ is tight. This proves $(1) \Rightarrow (4)$ and also yields (2) .

Conversely, a unit-norm equiangular tight frame meets the Welch equality, so its squared common angle is c . Consequently

$$H = (1 - c)I_M + c\mathbf{1}\mathbf{1}^*,$$

whose eigenvalues are $1 + (M-1)c = \beta$ once and $1 - c = \alpha$ with multiplicity $M-1$. Thus $(4) \Rightarrow (2)$, and $(2) \Rightarrow (1)$ is immediate.

Finally, the two universal inequalities give

$$B - A - (\beta - \alpha) = (B - \beta) + (\alpha - A) \geq 0.$$

Equality holds precisely when both nonnegative terms vanish, which is equivalent to (2) . This proves the equivalence with (3) . $\square$

4 Scope and edge cases

The theorem distinguishes the two one-sided questions. Equality in the *upper* bound alone characterizes all unit-norm tight frames, as already proved in [1]; it does not by itself force equiangularity. In contrast, equality in the *lower* bound already forces the complete equiangular tight structure. Since $\alpha > 0$, attaining it automatically makes H positive definite, so the Riesz assumption may equivalently be omitted from the hypothesis and recovered from the equality statement.

The restriction $M > N > 1$ is the redundant, nondegenerate regime of the source question. Existence of equality still depends on existence of an equiangular tight frame with parameters (N, M) ; when no such frame exists, the universal numerical bound is not attained.

5 Verification and provenance

The proof uses only the Gram identity from [1], the trace and Rayleigh quotient, the frame-potential identity, and equality in Cauchy–Schwarz. A constant-check script bundled with this packet verifies all rational identities over a broad finite parameter range. Cheap run indexes were searched for the exact title, authors, arXiv id, and the core phrases “outer-product Riesz bounds” and “equiangular outer products.” No later answer was found; the sole later parsed arXiv citation located was unrelated to this equality question. Novelty therefore remains subject to specialist review.

Reference. [1] P. G. Casazza, E. Pinkham, and B. Tuomanen, *Riesz outer product Hilbert space frames: quantitative bounds, topological properties, and full geometric characterization*, J. Math. Anal. Appl. 441 (2016), 475–498; arXiv:1410.7755.